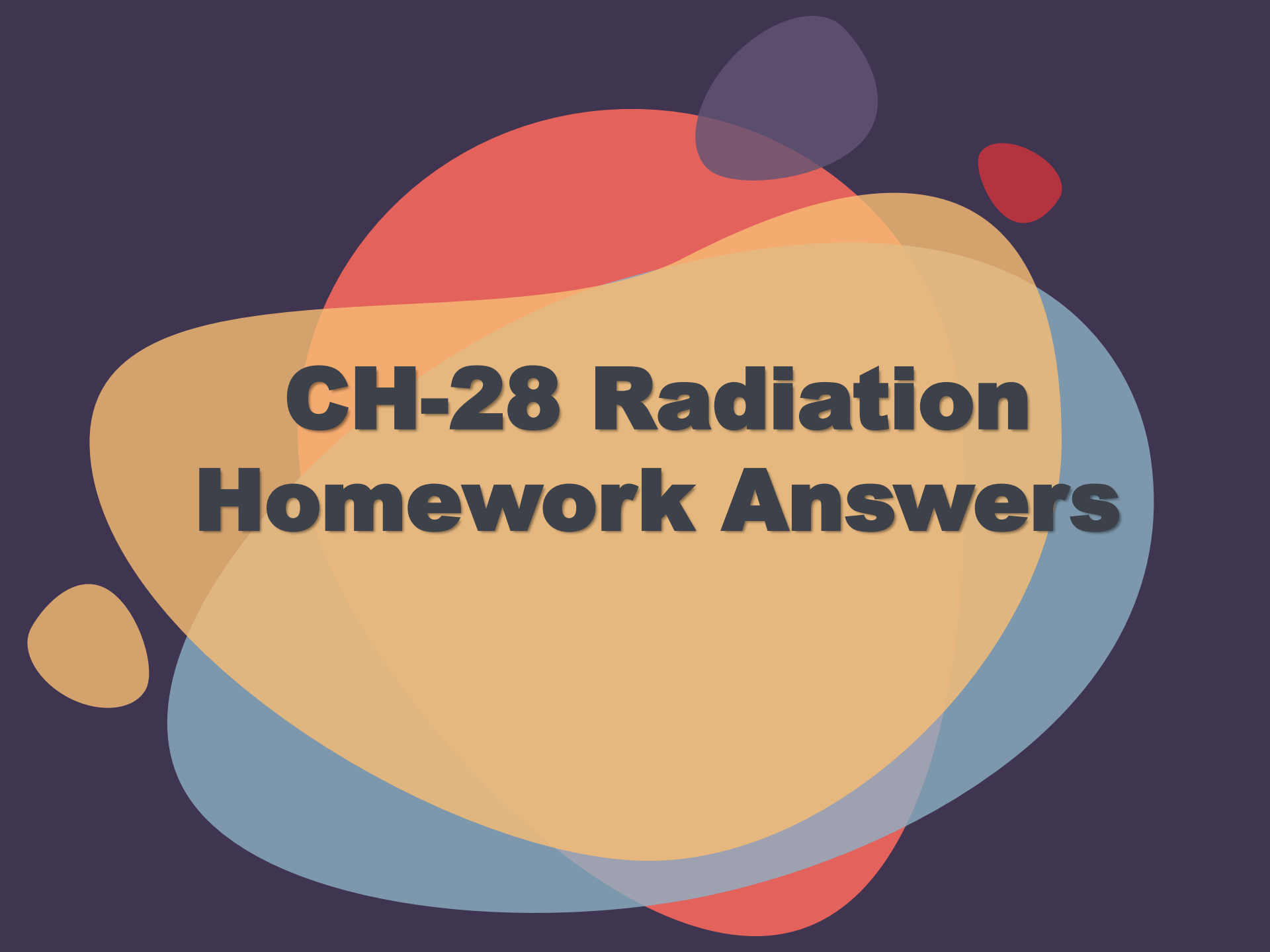




CH-28 Radiation Homework Answers



01

Which of the following organs and tissue is most sensitive to whole-body ionizing radiation?

A. Muscle tissue

B. Red bone marrow

C. Skin tissue

D. Lungs



02

The dose rate from an unshielded radiation source is 20 mrem/hour at a distance of 3 feet. What is the dose rate from this source at a distance of 15 feet?

A. **0.8 mrem/hour**

B. 4.0 mrem/hour

C. 10 mrem/hour

D. 500 mrem/hour

$$I_2 = I_1 \frac{(d_1)^2}{(d_2)^2}$$

$$I_2 = 20 \times \frac{3^2}{15^2} = 20 \times 0.04 = 0.8$$



03

Which type of radiation produces the most damage in a person when it interacts directly with living tissue?

A. Alpha

B. Beta

C. Gamma

D. It depends on the energy level.



04

What is the estimated radiation dose rate 1 foot from an unshielded 300 millicurie Mn-54 source? The energy level of the gamma photons is 0.835 MeV.

$$S = 6CE$$

A. 0.50 R/hour

B. 1.50 R/hour

$$S = 6 \times 300 \times 10^{-3} \times 0.835 = 1.503$$

C. 500 R/hour

D. 1500 R/hour



05

The statistically determined time interval for a radioactive isotope to decay by 50% is called the:

- A. LD50
- B. Decay rate
- C. Radiation emission rate
- D. Half-life of the material**



06

Which action most effectively minimizes radiation exposure?

A. Wearing a lead apron

B. Maximizing the distance from the radiation source

C. Minimizing the time spent near the radiation source

D. Taking iodine supplements before radiation exposure



07

Which device is best for measuring an employee's accumulated radiation dose?

- A. Proportional counter
- B. Scintillation counter
- C. Geiger counter
- D. Pocket dosimeter**



08

A curie is the amount of any radioactive isotope that will produce how many nuclear disintegrations per second?

A. 6.28×10^6

B. 3.7×10^6

C. 6.28×10^{10}

D. 3.7×10^{10}



09

Which is the most vulnerable to injury induced by laser energy?

A. Skin

B. Eye

C. Internal organs

D. Glands



10

The linear attenuation coefficient of lead for cesium-137 is 1.386 cm^{-1} . How thick must a lead shield be to lower the dose rate from 10 rem/hour to 0.1 millirem/hour?

$$I = I_0 e^{-\mu x}$$

A. 1.4 cm

B. 3.0 cm

C. 8.3 cm

D. 9.0 cm

$$0.1 \times 10^{-3} = 10 e^{-\mu x}$$

$$\frac{0.0001}{10} = e^{-1.386x}$$

$$\ln \frac{0.0001}{10} = -1.386x$$

$$x = \frac{-11.513}{-1.386} = 8.3 \text{ cm}$$